

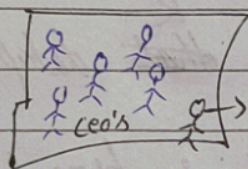
K.N.N

- ① KNN Classifier
- ② KNN Regressor

Before moving forward ~~we~~, this is one of the easiest algo. of ml.

Idea on which KNN Based on $\rightarrow$ you will become like the people so surrounded around you.

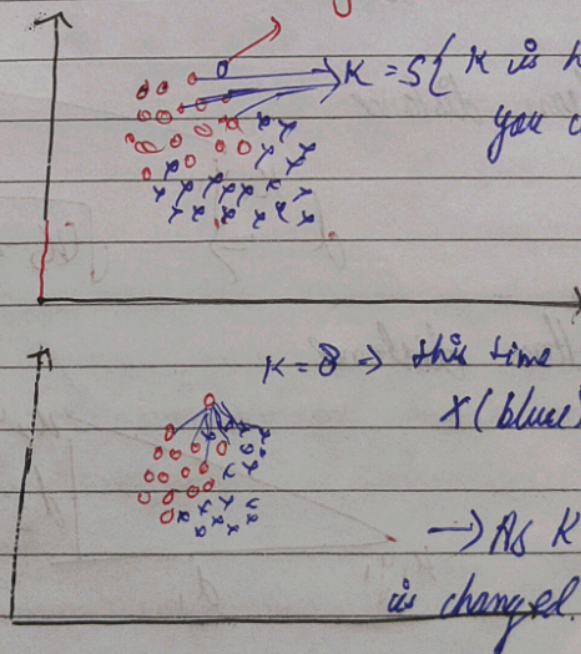
• if you are 5th on a group of smoker you will be 5th.



$\Rightarrow$ as all the people in room are CEO, a new person of room, $\rightarrow$ we will assume him to be a CEO also

• KNN classifier $\rightarrow$

f_1	f_2	f_3	y
-	-	-	0
-	-	-	1
-	-	-	0
-	-	-	0
-	-	-	1



majority is red(0) $\rightarrow$ it is assigned to red class.

$K=5$ (K is how many dp's you want to see)

$K=8 \rightarrow$ this time majority is x (blue) class

$\rightarrow$ As K increased class is changed.

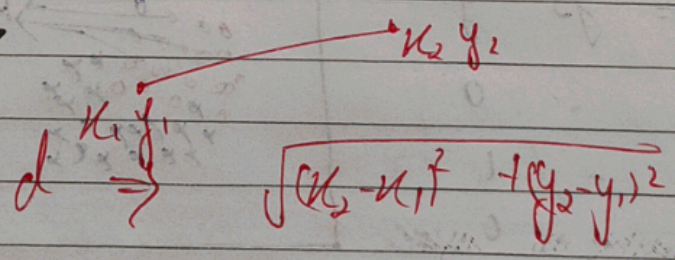
$K \rightarrow$ hyperparameter

Algo of Knn-classifier

- ① Plot the datapoint in n-d space
- ② initialise the k-value (no. of neighbour you want to consider)
 $\hookrightarrow$ k will always be greater than 0
 $\hookrightarrow$ generally $k \geq 3$
- ③ Calculate the distance of new dp wrt to all datapoints
- ④ sort the distance.
- ⑤ based on k find the class of ~~that~~ ^{k} datapoints
- ⑥ Take mode of all classes
- ⑦ assign the class of that which is present in majority.

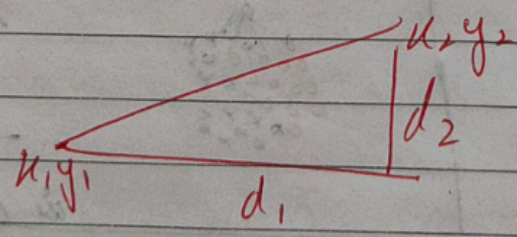
To calculate distance

- ① Euclidean distance



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- ② Manhattan distance

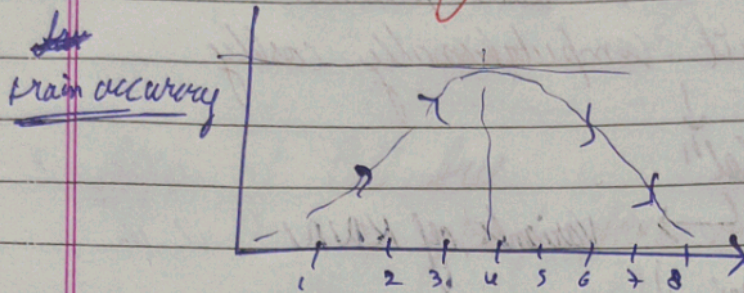


$$d = d_1 + d_2$$

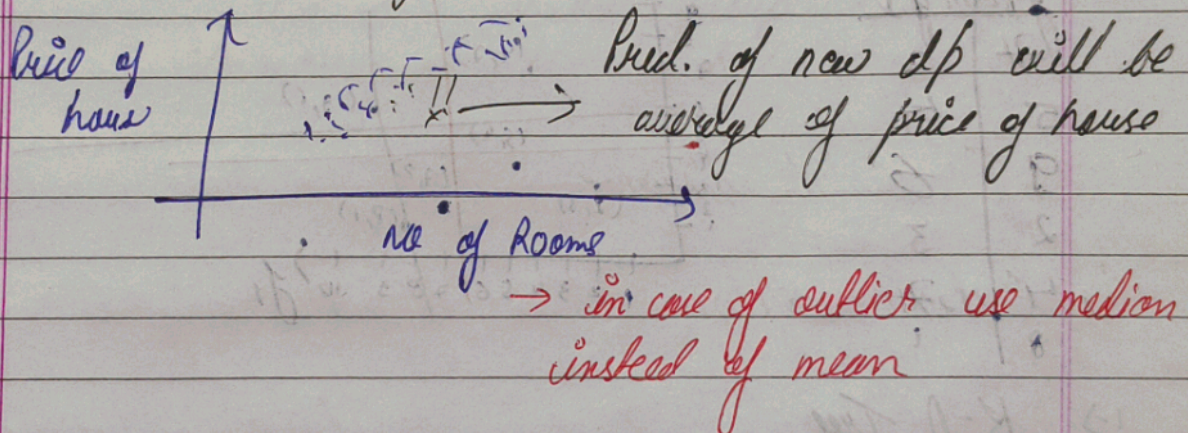
$$= (x_2 - x_1 + y_2 - y_1)$$

$K \Rightarrow$ a hyperparameter

$\rightarrow$ for K you keep the track of train/test accuracy.



• K.N.N Regressor



Regression algo

$\rightarrow$ All steps are same instead of mode use (mean or median) of K -nearest datapoints.

Variants of KNN

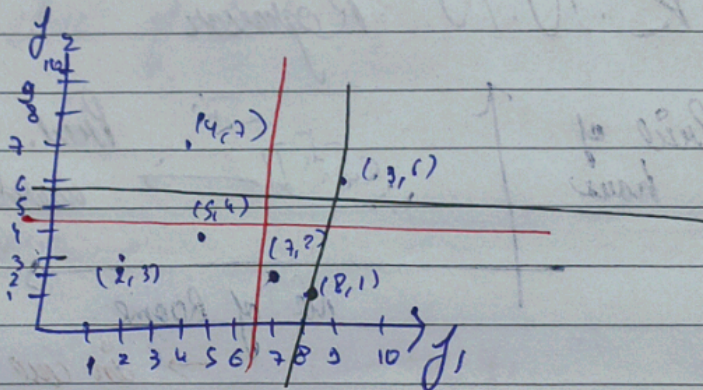
- till now we have seen that we are calculating the distance of $(n-1)$ data points.
- making it computationally costly.

Solⁿ

→ Variants of KNN

- 1) K-D Tree (K-dimensional tree)
- 2) Ball Tree

x_1	x_2
7	2
5	4
9	6
2	3
4	7
8	1



1) K-D Tree

→ idea → partition the data into binary tree, in order to implement idea follow steps below.

Step 1 → sort the features (x_1 & x_2)

$x_1 \rightarrow 2, 4, 5, 7, 8, 9$, $x_2 \rightarrow 1, 2, 3, 4, 6, 7$

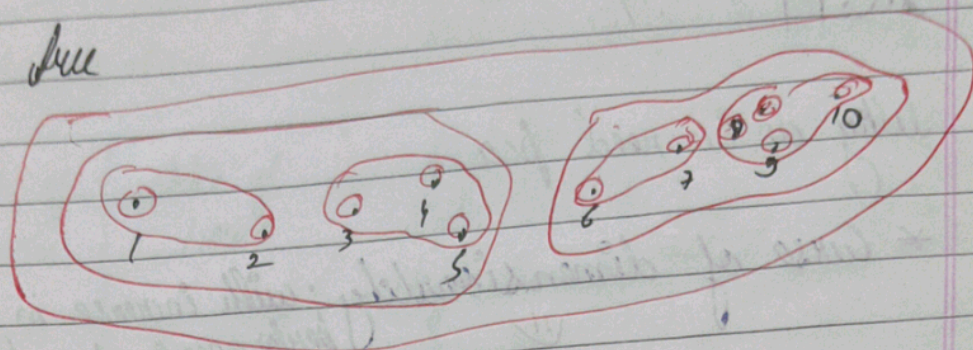
Step 2 → calculate the median of (x_1 & x_2)

$\frac{(5, 7)}{2} \rightarrow$ just take any of

Step 3 → partition the data based on median recursively.

Step 4 → make K-D tree → (Binary tree) → two child node

② Ball Tree



• Algo of Ball Tree $\hookrightarrow$ group the nearest data points recursively

- ① $\rightarrow$ each data point is ball in itself
- ② $\rightarrow$ keep combining till we have only one ball
- ③ $\rightarrow$ for test data $\rightarrow$ only need to calculate the distance
it's own ball/group to calculate
element or nearest ball element